

Roof Lifecycle Cost Analysis

Planned maintenance is projected to save the owner of a typical Hampton Roads school **\$1.34M** over the next 20 years.

About 35% less than reactive repair: \$990K lower spending plus \$352K more roof life at year 20. Range \$693K–\$1.75M from conservative to upside assumptions; \$698K in today's dollars at a 7% discount rate.

MARKET	PROPERTY	ROOF AREA	SYSTEM	AGE · CONDITION	NEW ROOF TODAY
Hampton Roads	K-12	120,000 sq ft	Modified bitumen	8 yrs · Good	\$960K

PAXSEAL NET COST

\$2.48M

\$2.39M spent + \$576K roof used – \$480K roof left

REACTIVE NET COST

\$3.83M

\$3.38M spent + \$576K roof used – \$128K roof left

REPAIR & STORM SAVINGS

\$1.63M

Unbudgeted repairs and storm damage avoided

ROOF PRICES

2× in 15 yrs

At 4.6%/yr, the long-run trend in commercial roofing costs

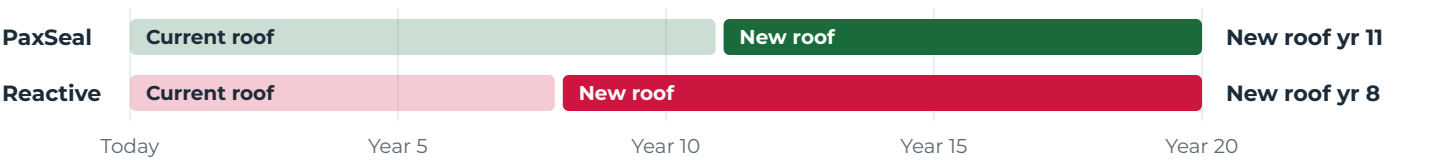
Cumulative roof spending, next 20 years

Nominal dollars. Vertical jumps are full roof replacements.



Roof life

When each scenario needs a new roof.



Where the savings come from

20-year totals

	PAXSEAL	REACTIVE	SAVINGS
PaxSeal fees	\$441K	—	-\$441K
Repairs paid by owner	\$234K	\$1.79M	+\$1.56M
Expected storm damage	\$140K	\$210K	+\$70K
Roof replacements	\$1.57M	\$1.38M	-\$190K
Cash spending	\$2.39M	\$3.38M	+\$990K
Current roof value used	\$576K	\$576K	—
Roof life left at year 20	-\$480K	-\$128K	+\$352K
Net 20-year cost	\$2.48M	\$3.83M	+\$1.35M

What this means

- 1 A later replacement. The next new roof moves from year 8 to year 11, leaving 10 years of roof life at year 20 instead of 2.
- 2 Smaller repairs. Emergency repairs run about 3.9× the planned rate in a school, and the PaxSeal fee covers half of planned repairs. Together that saves \$1.56M.
- 3 A predictable budget line. Emergency repairs become a fixed annual cost the board can plan for, and leaks stop closing classrooms.
- 4 The program pays for itself. Each \$1 of fees avoids about \$3.25 of other cash roof costs.

Local risk profile

Norfolk city, VA (51710)

Hurricane / tropical-storm wind

About 4.8 tropical storms or stronger within 50 nautical miles of Norfolk per 20 years (NOAA HURDAT2 1975–2024). Expected storm damage to this roof over 20 years is \$140K if maintained and \$210K with reactive repair only, before insurance. A documented maintenance history also helps when an insurer reviews a claim.

Hail	Relatively Low	■ ■ ■ ■ ■
Strong wind	Relatively Moderate	■ ■ ■ ■ ■
Tornado	Relatively Moderate	■ ■ ■ ■ ■
Hurricane	Relatively Moderate	■ ■ ■ ■ ■
Winter weather	Relatively Moderate	■ ■ ■ ■ ■
Ice storm	Relatively Moderate	■ ■ ■ ■ ■
FEMA National Risk Index rating		

Key assumptions

The basis for each figure in this analysis.

ASSUMPTION	VALUE	BASIS
Roof life	20 / 15 yrs	Maintained / reactive. Industry life tables; maintenance adds ~5 yrs (NRC Canada, warranties)
Roof cost growth	4.6%/yr	BLS Producer Price Index, nonresidential roofing, 2007–2025
Replacement cost	\$8.00/sq ft	Hampton Roads contractor price ranges
Reactive repair cost	3× planned × 1.3	Industry estimates; ×1.3 for leak damage (school). PaxSeal fee covers half of planned repairs
Storm damage	0.44%/yr	Of roof cost, before insurance; ×1.5 if reactive. FEMA National Risk Index
PaxSeal fee	\$16,400/yr	Year 1, rising 3%/yr. Estimate; your proposal states the actual fee

Next step: a roof assessment and a written PaxSeal proposal for this building. Call PAX Services Group at 1-855-729-7510 or email info@paxserv.com.

Estimates, not a quote. Projections from Hampton Roads market research and industry data, in nominal dollars. The PaxSeal fee is an estimate; your proposal states the actual price. Net cost includes wear on the current roof, less roof life left at year 20; both roof-value figures are in today's dollars. Storm damage is before insurance.